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Bi-Weekly Report of (Team# 1 - JAVES AI)
Week #6

What were the goals for the last 2 weeks?

Junwoo Lee

1. Clarify the workflow including authentication, case creation, candidate review and more.
2. Develop a Streamlit interface for uploading a reference image / entering the last known camera zone and search time range.
3. Build an interactive prototype so the team can review the interface before backend integration.

Arystan Kamalov

1. Multi-camera input: Implement four synchronized camera feeds at 1280×720 resolution, with matching frame IDs, timestamps, and ground-truth annotations.
2. Simulation: Create a controlled environment with three autonomous pedestrians, obstacles, and overlapping camera coverage to test movement, occlusion, and camera transitions.
3. Detection and tracking: Integrate YOLO26, per-camera BoT-SORT, and OSNet person re-identification into one pipeline that produces local tracks and cross-camera identity assignments.

Viktor Korotkov

1. Build prototype of Routing worker (core, with physically possible route hypothesis)
2. Verify Routing through tests

Sihyun Jeong

1. Understand the clip and retention worker requirements.
2. Plan a prototype for clip extraction, merging, and data cleanup.

What goals were accomplished this week?

Jungwoo Lee

1. Refined the interface requirements with team members through discussion and updated the workflow. Updated part: operators retrieve cases by case ID rather than providing a separated search history.
2. Built a prototype login screen with ID, password blanks.
3. Added a case-creation form with reference image upload, camera-zone selection, time inputs and a timeout.
4. Implemented candidate-review features; confirmation, removal, recalculation requests.
5. Added a camera-zone route display, clip playback, case saving, retrieval, deletion.

Arystan Kamalov

1. Completed the simulation environment with three autonomous pedestrians, obstacles, and four synchronized cameras.
2. Integrated YOLO26 detection, BoT-SORT tracking, and OSNet re-identification into one pipeline. The system now generates local tracks and cross-camera identity assignments, although identity fragmentation still requires improvement

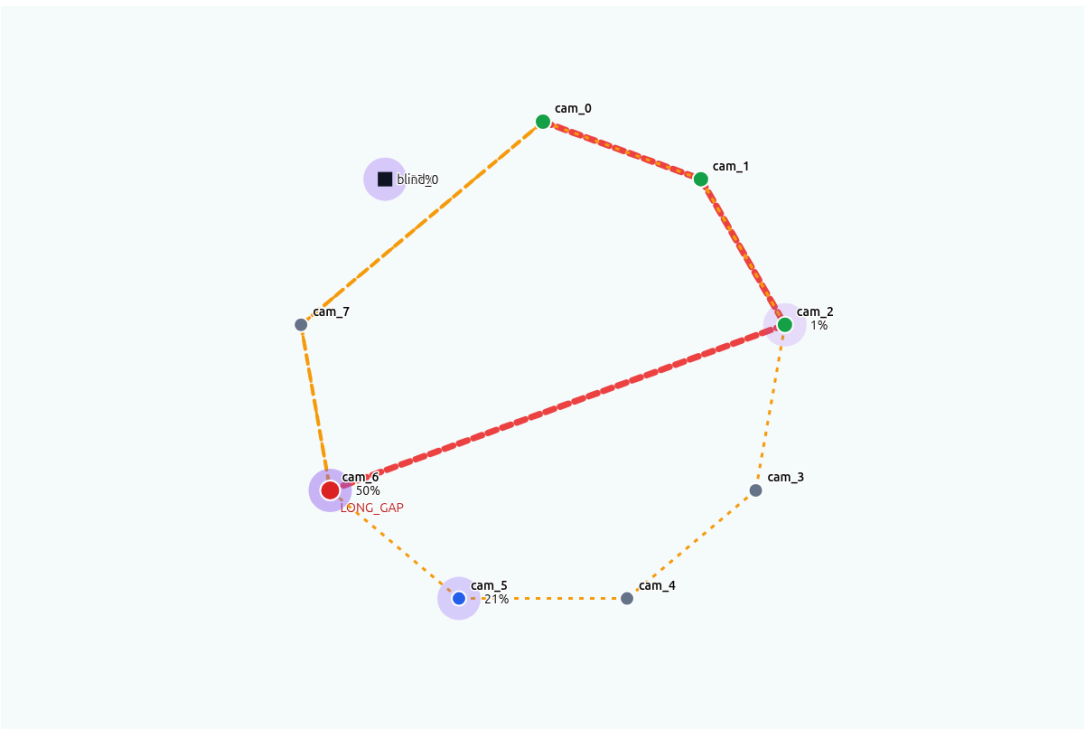
Viktor Korotkov

1. Implemented pure init_state() and step(), SiteGraph validation, time gates, hypothesis branching, scoring and operator confirmation/rejection.
2. Implemented DONE-empty handling, retries, unobserved windows, OPEN upserts, dormant searches and LONG_GAP confirmation.
3. Implemented all Routing modes, stop conditions and deterministic multi-floor SVG rendering.

```
plugins: anyio-4.10.0
collected 10 items

tests/test_algorithm.py::test_1_gate PASSED                    [ 10%]
tests/test_algorithm.py::test_2_ring_chord PASSED              [ 20%]
tests/test_algorithm.py::test_3_empty_probe PASSED             [ 30%]
tests/test_algorithm.py::test_4_lateness PASSED                [ 40%]
tests/test_algorithm.py::test_5_re_entry PASSED                [ 50%]
tests/test_algorithm.py::test_6_operator PASSED                [ 60%]
tests/test_algorithm.py::test_7_modes_and_stop PASSED          [ 70%]
tests/test_algorithm.py::test_8_baseline PASSED                [ 80%]
tests/test_algorithm.py::test_9_purity PASSED                  [ 90%]
tests/test_algorithm.py::test_10_efficiency PASSED             [100%]

========================== 10 passed in 0.32s ==========================
(base) viktor@dark-8yte:~/Downloads/JAVES_AI_project$
```



Sihyun Jeong

1. Identified my worker responsibilities and their connections to shared storage and the API.
2. Outlined the clip-processing and retention workflows based on the Miro requirements.
3. Selected the shared sample video for initial testing.

Reflect critically on any goals not accomplished.

Jungwoo Lee

The project has not yet been connected to the actual search operation. Integration requires agreement on the case and candidate data formats, etc. I'll discuss these interfaces with the team members before replacing the components.

Arystan Kamalov

1. Completed the basic simulation, but its ground-truth annotations do not measure whether a person is actually visible behind obstacles. The initial implementation uses projected body bounds, which simplified development but limits detection evaluation
2. I did not achieve consistently stable identities across cameras. The pipeline runs, but occlusion, synthetic appearance, and inaccurate position estimates contribute to identity fragmentation. Integration took priority over systematic tuning

Viktor Korotkov

1. Routing has not been validated on real Perception output, because the real worker is not available now (in progress, Arystan is working on it).
2. The difficult mismatch benchmark reaches only 43% recall.

```
found_rate=0.430 smart_camera_minutes mean=13.617 median=14.500 p90=14.500
```

Sihyun Jeong

Implementation has not started yet, as I focused on understanding the requirements and learning the tools. I will begin with a standalone clip-processing prototype and clarify the clip states and retention timing with the team.

What are the goals for next two weeks?

Jungwoo Lee

1. Agree with backend developers on the required fields(cases, candidates, video clips…)
2. Connect the dashboard to real candidate-result; displaying camera IDs, timestamps..
3. Connect confirmation, removal and recalculation actions to the agreed backend interface.

Arystan Kamalov

1. I will create three reproducible test scenarios covering clear visibility, occlusion, and camera transitions, with at least 30 seconds of synchronized footage each. I will add visibility-aware annotations and verify them on 100 sampled person observations
2. Establishing baseline IDF1 and identity-switch counts on these scenarios, then tune cross-camera association. My target is to reduce identity switches by at least 20% without lowering IDF1 on the same recordings

Viktor Korotkov

1. Validate real Perception payloads for success, empty, failure, unavailable and OPEN cases.
2. Validate route rendering and travel-time models on real building maps.

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Sihyun Jeong

1. Implement and test clip extraction and merging using FFmpeg.
2. Add clip-status management and basic retention cleanup with audit logs.
3. Agree with teammates on the data formats and API responsibilities.

How many hours were spent on each goal noted above?

Jungwoo Lee

1. Requirements & workflow definition(2h)
2. Streamlit learning and interface development(3h)
3. Prototype integration(2h)
4. Testing(1h)

Arystan Kamalov

1. Simulation: 5 hours
2. Detection and Tracking: 3 hours
3. Calibration and Visualization: 2 hours

Viktor Korotkov

1. Routing core (8h)
2. Tests (1h)

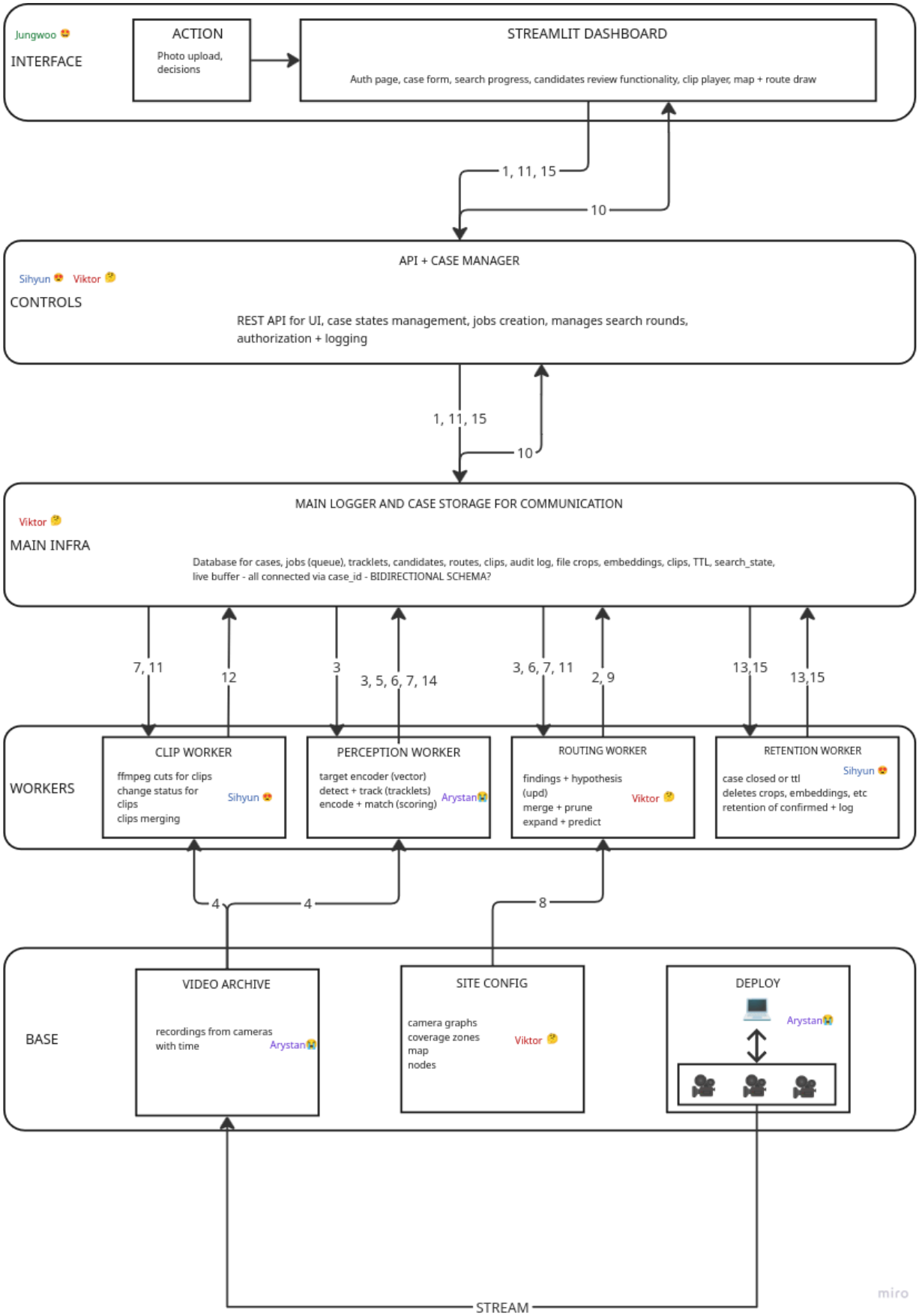
Sihyun Jeong

1. Requirements: 1h
2. Workflow: 0.5h
3. Planning: 0.5h
4. Troubleshooting: 2h

Additional note: Goal achieved with the whole team

As a team, we discussed and planned the complete project architecture (detailed, beyond the things presented for the first time), including component responsibilities, data flows, worker interfaces, Main Infrastructure, Controls, Dashboard, authentication, storage, deployment and the integration path from the MVP to the complete product. We are trying to stick to AGILE principles. Here is the architecture (module-monolitic):

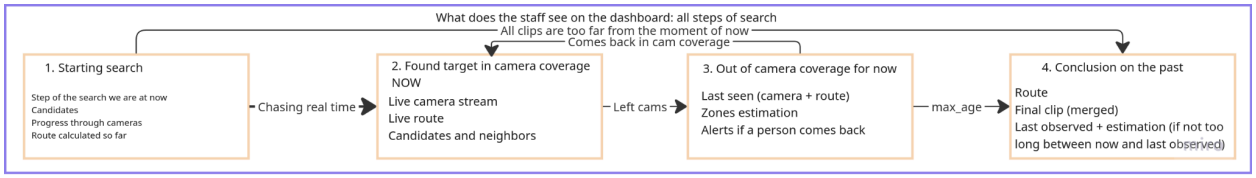
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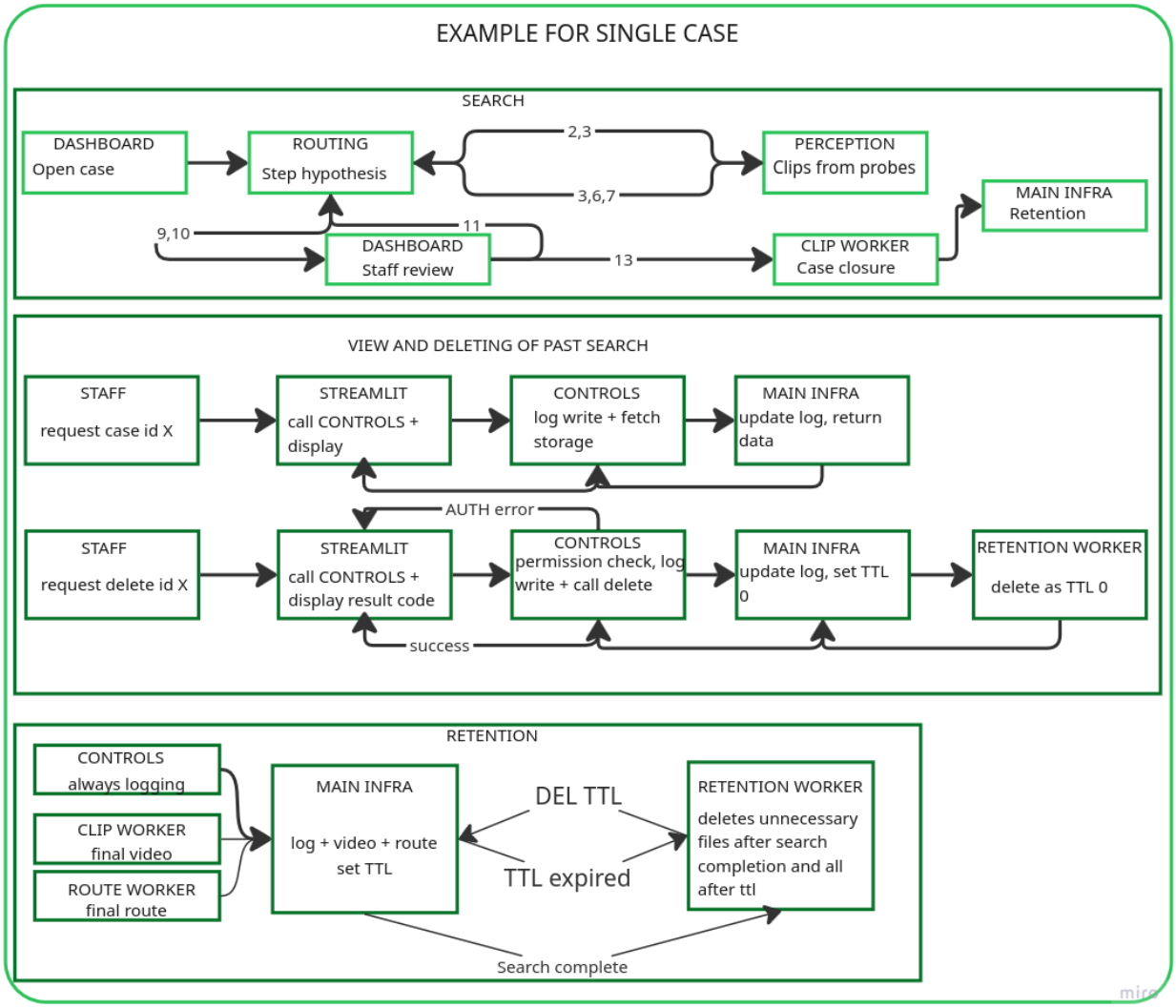


Capstone Design

Weekly Report # 1

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COMMUNICATION

1. Case request (reference image, last known {camera | zone, time}, window {start, end}, operator_id fields) - when creating case
2. Probe request ({probes[{probe_id, camera_id, t_from, t_to | OPEN, priority, reason, hypothesis_ids[]}], mode, done, stop_reason} fields) - probes for each camera step t_to = OPEN is live cam subscription
3. Probe job ({job_id, probe_id, kind: TARGET | PROBE, camera_id, t_from, t_to | OPEN} fields) - answer like status: DONE | FAILED | TIMEOUT, result {n_tracklets, n_above_floor, best_score}
4. Frames (get_frames(camera_id, t_from, t_to) -> frames, absolute time ; and get_segments(...) gives file for clip now() to get source time)
5. Target embedding ({case_id, vector[512], source: reference | confirmed_tracklet, crop_path, quality} fields) - is a gallery inside each case (tracklets added after staff confirmation)
6. Tracklet ({tracklet_id, camera_id, t_start, t_end, open, clipped_start, clipped_end, bbox_first, bbox_last, embedding, crop_paths[], clip_ref} fields) - open means track ongoing
7. Candidate ({candidate_id, tracklet_id, appearance_score ∈ [0, 1], cosine}) - routing adds reject reason and final score
8. Site graph (cameras[], regions[], edges[] {id, t_min, t_typ, t_max, d_m}, blind_nodes[] {dwell}, outside {entrances{camera: p}, dwell}, map, clock_offset_s) for yaml config
9. Hypothesis + estimate (hypotheses[] {id, visits[], gap, score, status: ALIVE | DORMANT | ENDED}, best, ambiguities[], estimate {zones[{zone, p}], valid_until}, mode) - search_state read only by Routing worker
10. Case view - All the dashboard needs: mode, search progress, candidates (thumbnail, score, status), best hypothesis & alternatives, estimate, live {camera_id, stream_url, bbox, track_age_s}, watch[], clips. UI reads only this. 2 updates per second in search mode.
11. Review desicion (candidate_id, decision: confirm | reject, operator_id (staff), timestamp fields) - to change clip state and recalculate route fields)
12. Clip record ({clip_id, candidate_id, camera_id, t_from, t_to, state: HOLD | CONFIRMED | RELEASED, file}. export_merged(case_id)) - merges clips with export_merged(case_id)
13. Case closure ({case_id, closed_by}) - delete all except final route, confirmed merged clip + log
14.Live feed - feed with drawn target box in live-buffer (files/live/{case_id}/{camera_id}/, TTL secs) and line live_state {camera_id, track_id, bbox, age_s}. API gives buffer as MJPEG or HLS; in CaseView that is stream_url.
15.State change (SAVE | DELETE case, case_id, operator_id, timestamp)

miro

Since everybody participated in designing the complete architecture, we cannot divide the job according to the conditions. Time spent is approximately 3 hours, including questions, reconsiderations, etc.